SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – COMPARITIVE ANALYSIS

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO4	**Assessment:**	AT1 - COMPARITIVE ANALYSIS
Weightage:	30 % of CIA	**Total Marks:**	100
CO4 Statement:	*Identify the components and structure of mobile application for Android and Windows OS-based devices.*		
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Section / Batch:	B.TECH (IT) / 2024	**Date:**	18/08/2026

Code of Conduct

I ___Gowtham Reddy(192372158)_______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _igowthamreddy_______

Instructions

- Duration: 60 minutes
- Number of questions : 01

Faculty In-charge	Course Coordinator	Head of Department
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

Annexure A - QUESTIONS

1. A telecommunications company wants to develop a simple wireless information service for low-resource mobile devices. Compare WML-based applications and J2ME/JME applications in terms of architecture, execution environment, device capabilities, and development requirements. Analyze which technology is better suited for the proposed service.

Aspect	WML-Based Applications	J2ME/JME Applications
Architecture	Uses a client-server architecture. Content is stored on a web server and delivered through a WAP gateway to the mobile device.	Uses a Java-based application architecture. Applications run directly on the mobile device using the J2ME runtime environment.
Execution Environment	Runs inside a WAP/WML browser. The device mainly requests and displays WML pages.	Runs inside a J2ME virtual machine (KVM) using configurations such as CLDC and profiles such as MIDP.
Device Capabilities	Designed for very low-resource devices with limited memory, processing power, and small screens. Mainly supports text and simple interaction.	Supports richer device functionality, including graphics, user input, local storage, networking, multimedia, and device APIs, depending on the handset.
Network Dependency	Generally highly dependent on network connectivity because content is retrieved from the server.	Can support local processing and storage, reducing the need for continuous server communication.
User Interface	Simple, mainly text-based and card/deck-oriented interfaces.	Provides richer interactive graphical interfaces, forms, menus, games, and custom screens.
Development Requirements	Requires knowledge of WML, WAP, XML-like markup, and server-side technologies.	Requires Java programming, J2ME APIs, CLDC/MIDP knowledge, and suitable Java development tools.
Processing	Most processing is performed on the server; the device primarily displays the received content.	More processing can be performed on the mobile device itself.
Application Complexity	Suitable for simple information services, such as news, weather, stock information, and basic queries.	Suitable for interactive and feature-rich applications, including games, enterprise applications, and offline-capable utilities.

Resource Usage	**Very low, making it suitable for older low-resource mobile devices.**	**Generally higher because Java applications require a JVM/runtime and additional device resources.**

1. Architecture

WML-based applications: WML applications follow a client-server architecture. The information is stored on a web server and delivered to the mobile device through a WAP gateway. The mobile device mainly requests and displays the information.

J2ME/JME applications: J2ME applications use a Java-based architecture in which the application is installed and executed on the mobile device using a Java Virtual Machine (JVM).

2. Execution Environment

WML: WML applications execute through a WAP/WML browser. The device does not need to run a complete application locally.

J2ME/JME: J2ME applications execute directly on the device using a J2ME runtime environment, commonly based on CLDC and MIDP.

3. Device Capabilities

WML: WML is designed for low-resource mobile devices with limited memory, processing power, small displays, and basic input facilities. It mainly supports text and simple interactions.

J2ME/JME: J2ME provides more advanced capabilities such as graphics, local storage, networking, multimedia, and interactive interfaces, depending on the device.

4. Development Requirements

WML: Development requires knowledge of WML, WAP, XML-based markup, and server-side technologies. It is relatively simple for basic information services.

J2ME/JME: Development requires knowledge of Java programming, J2ME APIs, CLDC, MIDP, and Java development tools. It is more suitable for complex applications.

5. Suitability for Proposed Service

For a simple wireless information service for low-resource mobile devices, WML is more suitable. It requires fewer device resources and is effective for services such as news, weather, alerts, and basic information retrieval. J2ME/JME would be more suitable if the service required advanced graphics, multimedia, offline processing, or complex interaction.

6. Conclusion

WML is the better choice for the proposed service because the main requirement is a simple, lightweight service for low-resource devices. J2ME/JME is preferable when richer and more interactive mobile applications are required.